

Botond Kovács

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Introduction

In my data analysis project, I focused on the concept of personalized education, specifically, on Deep Knowledge Tracing.

- Personalized education: tailoring educational content and strategies to individual students' needs and learning styles using data-driven algorithms and models.
- Deep Knowledge Tracing: using deep learning to model personalized education.

My initial goals were to implement the models from the paper, modify them to improve AUC, and extract the latent structure in the data using a novel approach.

Data

I focused on one dataset, the ASSISTments 2009-2010 dataset. It consists of online learning sequences that track the progress of students as they answer different questions. To each student a sequence belongss, with each data point representing a question attempt. Key attributes:

- timestamp of the attempt
- question ID
- correctness = whether the student answered correctly
- skill ID

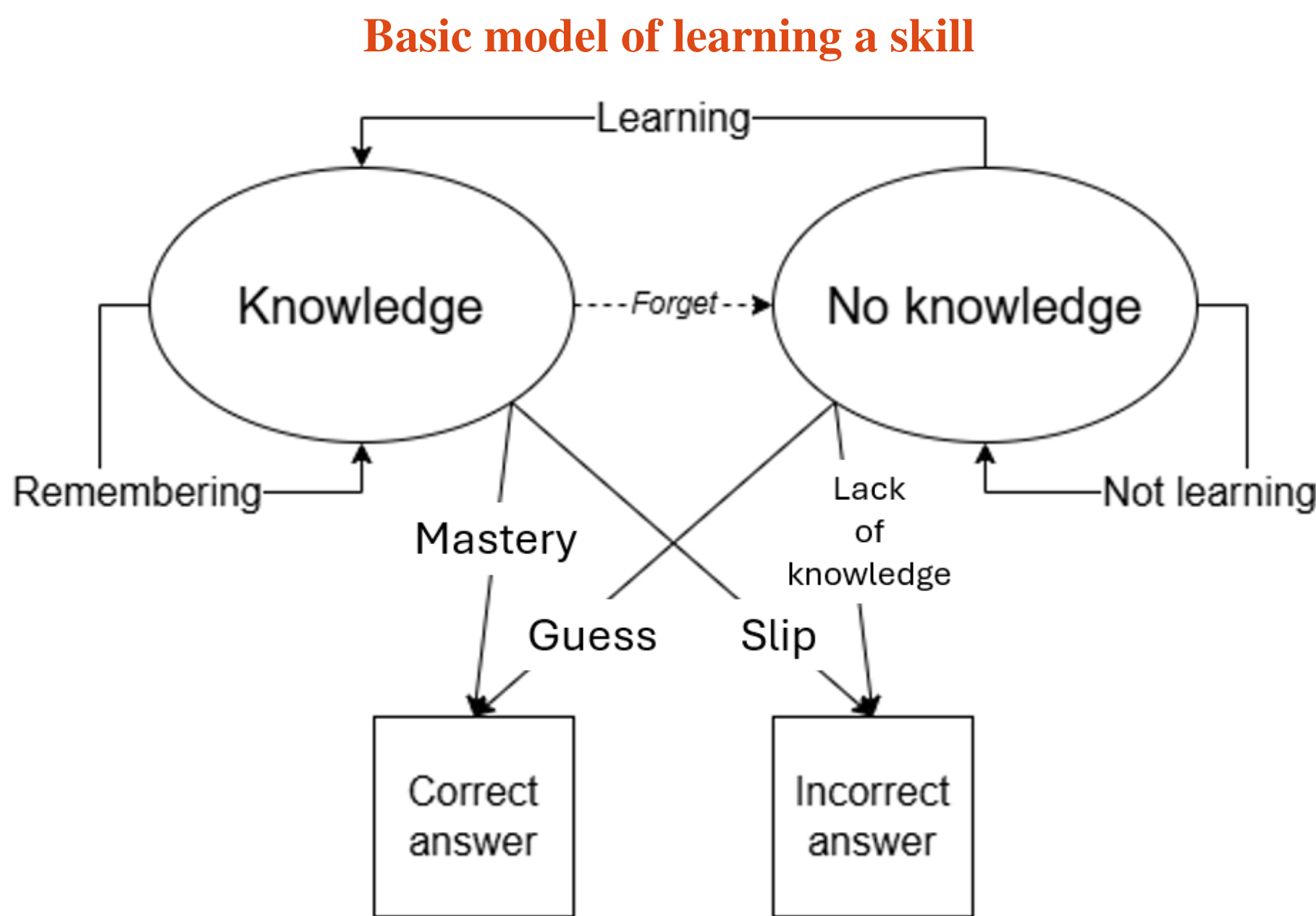
Dataset Overview

#Users	#Skills	#Answers
3 657	109	301 088

Bayesian Knowledge Tracing (BKT)

The basis of Deep Knowledge Tracing is Bayesian Knowledge Tracing.

- Bayesian Knowledge Tracing: modeling learning a skill while giving answers to questions with a Hidden Markov model
- Flaws: assumes that skills are independent



BKT Experimental Results

Model	Benchmark	Implemented BKT	BKT with Forget
AUC	0.67	0.67	0.70

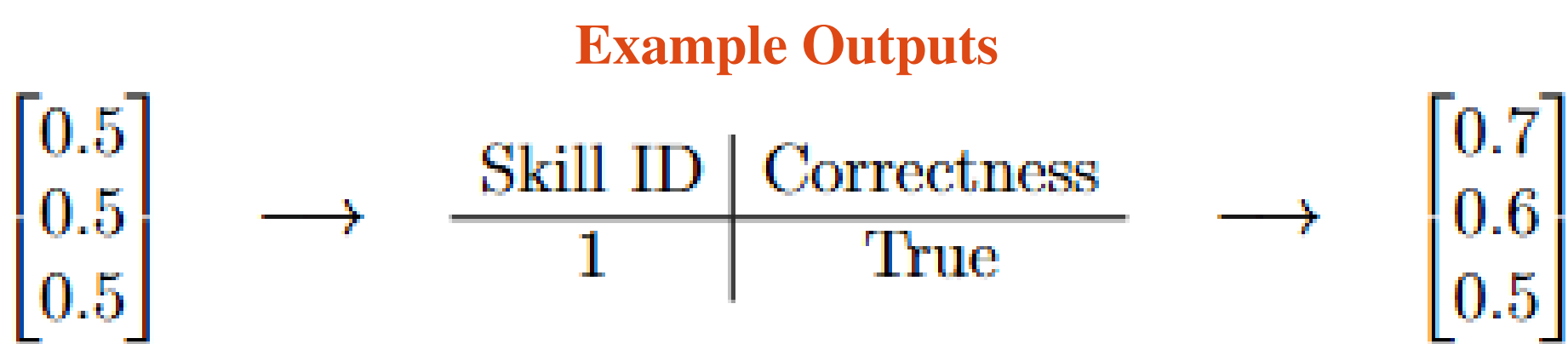
Deep Knowledge Tracing (DKT)

DKT = BKT - HMMs + RNNs

It allows us to use a single model for various skills while incorporating more complex connections.

Inputs and Outputs

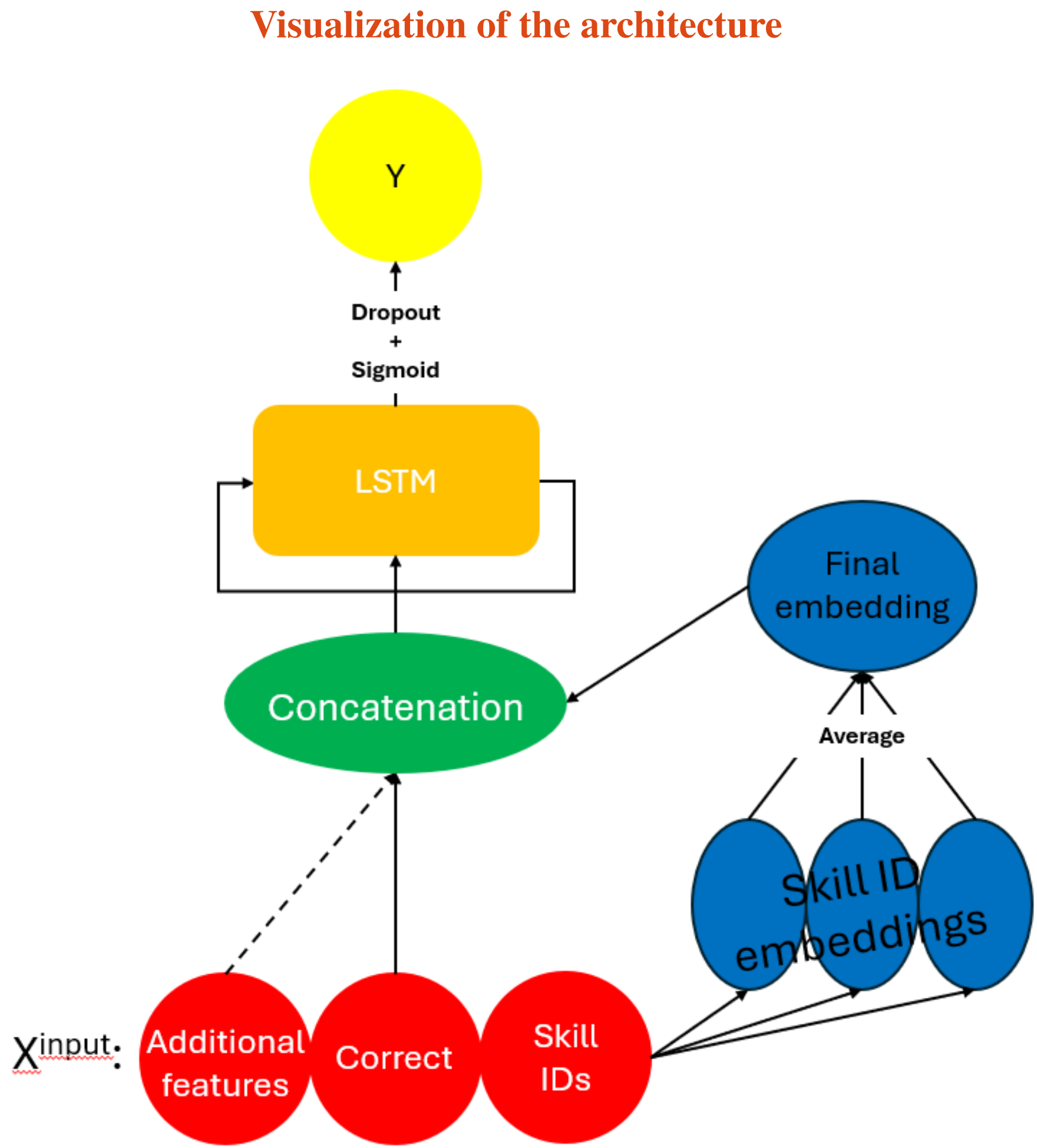
- Input: a sequence of answers, where each answer is paired with its corresponding skill ID, correctness, and any additional available features.
- Output: a vector for each item in the sequence with size equal to the number of skills in the data, with each coordinate representing the probability that the user would correctly answer a question for a given skill.



Model

The architecture is based on an LSTM model, which receives the concatenation of the following:

- additional features (if any)
- correctness
- average of the embeddings of the corresponding skills



DKT Experimental Results

	Benchmark	Simple	Complex
Train Set AUC		0.799	0.816
Test Set AUC	0.86	0.794	0.802

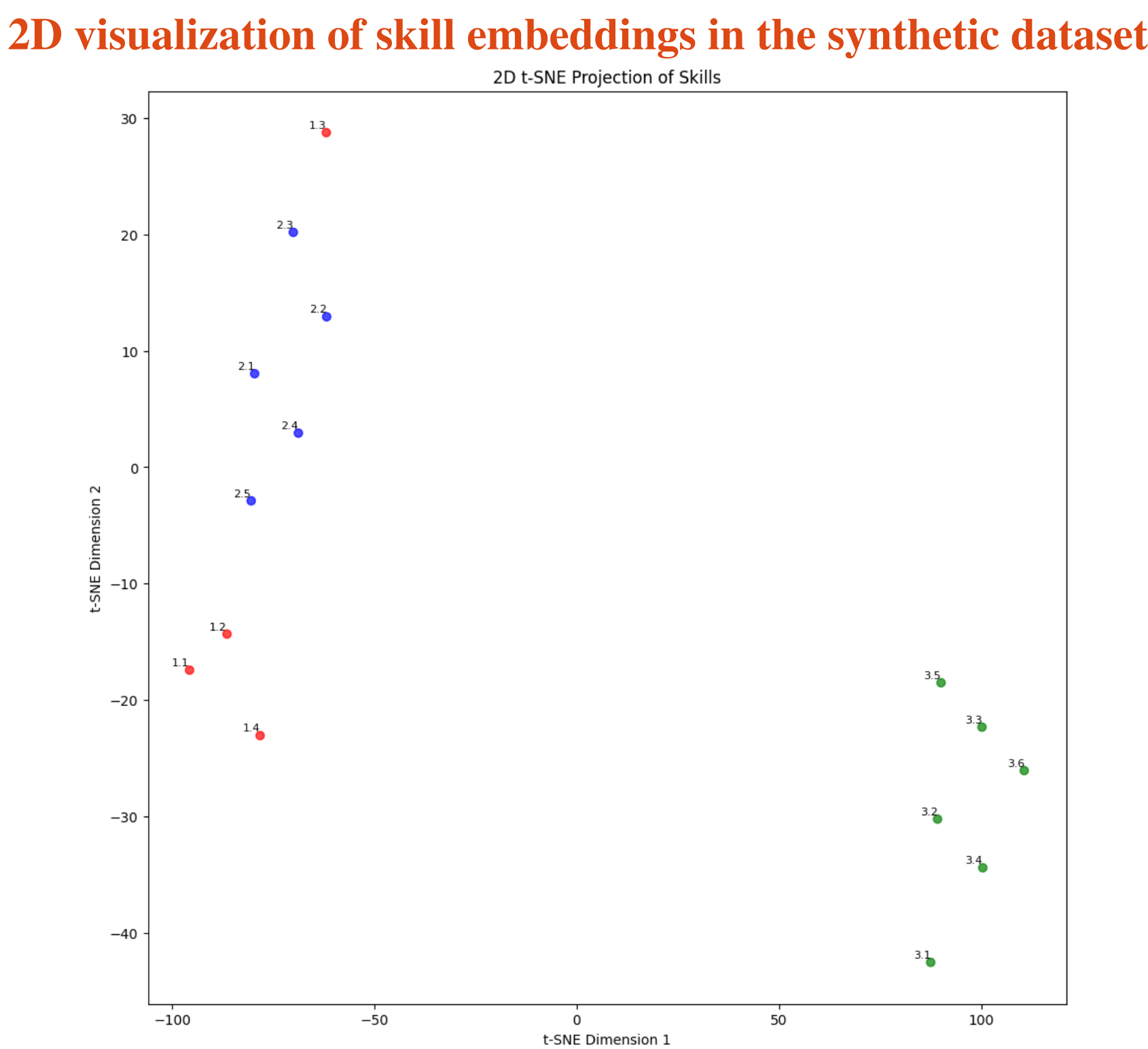
Latent Structure

Given that the model employs embeddings for skills, it was hypothesized that similar skills → similar embedding values

This would enable the construction of a structured relationship between different skills, allowing for groupings such as geometry and algebra.

Despite testing multiple trained models and various similarity metrics, the hypothesis did not hold for our real-world dataset. In each case, similar skills resulted in random embedding values.

However, when applied to a synthetic dataset, the model successfully identified similar skills with high accuracy. This suggests that the method has potential.



Future Work

- Trying out the xLSTM architecture to potentially enhance model performance and reveal latent structures.
- Application of explainability techniques, like SHAP values.
- Experimenting with the proposed structure detection method via embeddings on alternative datasets.

Conclusion

These findings highlight the challenges in accurately modeling student learning and emphasize the need for further experimentation.

References

[1] M. Beck, K. Pöppel, M. Spanring, A. Auer, O. Prudnikova, M. Kopp, G. Klambauer, J. Brandstetter, and S. Hochreiter. xlstm: Extended long short-term memory.

[2] A. T. Corbett and J. R. Anderson. Knowledge tracing: Modeling the acquisition of procedural knowledge.

[3] C. Piech, R. S. J. Baker, D. J. Bernstein, J. Cescovini, and S. Wong. Deep knowledge tracing.